

12. Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	SWTID-2026-4570
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space PEP-8 style throughout
2	Proper File Structure	Files and folders are logically organized	Yes	Feature-per-module (qna.py, quiz_module.py, ...) plus utils/, models/, templates/, static/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	e.g. is_gemini_active, best_model_name, explain_tokenizer
4	Function / Method Names	Functions are descriptively named	Yes	e.g. get_chat_response(), call_gemini_with_retry(), init_local_t5()
5	Code Comments	Inline and block comments explain logic	Partial	Docstrings present on most functions; inline comments are sparse in places
6	Modular Design	Code is split into reusable functions/modules	Yes	Shared Gemini/T5 logic centralized in utils/gemini_service.py and utils/local_t5_service.py
7	No Redundant Code	Duplicate or unused code is removed	Partial	main.py exposes both router-based (/api/...) and direct (/qa, /quiz, ...) endpoints for the same features — a known duplication to refactor
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/except with logging around every Gemini/T5 call and DB write, with mock-data fallback

Reusable Components / Modules

S.No	Component / Module	Language / Technology	Where Reused	Reusability
1	call_gemini_with_retry()	Python (utils/gemini_service.py)	Used indirectly by every get_*(response builder in gemini_service.py	High
2	log_query() / log_ai_response()	Python (utils/db_manager.py)	Called from every router (qna, explanation, quiz, summary, learning_path) and from main.py's direct endpoints	High
3	get_base_context()	Python (main.py)	Shared by all HTML page routes to pass gemini_configured to templates	Medium
4	load_t5_pipeline()	Python (utils/local_t5_service.py)	Warm-started at app startup and reusable by any future local-model feature	Medium
5	clean_json_block()	Python (quiz_module.py)	Strips markdown fences from LLM output before JSON parsing	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear feature-per-module layout; router/direct-endpoint duplication is the main deduction
Readability	4	Descriptive names and docstrings throughout
Reusability	4	Shared service layer (gemini_service.py, db_manager.py) is cleanly reused across all five features
Documentation / Comments	3	Good docstrings; README and workflow.md are thorough, but inline comments are inconsistent
Overall Score	4	Solid, demo-ready codebase; recommended next step is consolidating the duplicate endpoint surface